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Congenital diaphragmatic hernia (CDH) mortality without surgical repair? A plea to clarify surgical ineligibility

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Abstract
Purpose: Little is known about liveborn CDH patients who die without surgery. We audited a national CDH cohort to determine whether these patients were different from patients who received CDH repair.
Methods: A national CDH database was analyzed (2005–2009). After excluding infants with severe physiologic instability and genetic/congenital malformations, a potential surgical candidate (PSC) subgroup was identified. PSCs were compared to the operative group (OG) and the operative non-survivor (ONS) subgroup. Standard statistical analyses were performed.
Results: Of 275 liveborns, 35 (13%) died without surgery. The PSC subgroup ($n=11$) had a median survival of 10 days (range: 3–18). Ten of 11 PSC infants were treated in ECMO centers, with 4 receiving ECMO. No differences in BW, GA, and rates of minor malformation were observed between PSC and OG patients. While neonatal illness severity (SNAP-II) predicted overall mortality, SNAP-II scores were similar between PSC and ONS groups (34 vs. 29; $p=0.431$). Furthermore, greater than 80% of infants with SNAP-II scores between 30 and 39 survived in the OG cohort.
Conclusion: Our analysis demonstrated that PSCs were similar to infants offered surgery based on illness severity and the presence of congenital malformations. We suggest that criteria for surgical ineligibility be developed to standardize the selection of surgical candidates.

While population-based estimates of survival for liveborn CDH infants who receive surgery approaches 80% [1], there are a small but significant number of infants who do not survive to surgical repair. While the reasons for surgical ineligibility are generally related to severe physiologic instability and lethal genetic/congenital anomalies [2], very little is known about this specific cohort of patients.

Data from CAPSNet have recently shown that of the 51 infants with CDH who died after admission to a CAPSNet center between 2005 and 2010, 35 never received surgical repair [1]. Given that significant institutional practice pattern and outcome variation exist across Canada with respect to the perinatal management of infants with CDH [3], we sought to analyze infants who died prior to surgery to (a)

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document and clarify the reasons for surgical ineligibility and (b) to potentially identify a subset of patients who may have been considered potential surgical candidates (PSC), by comparing demographic and physiologic data between infants who did and did not receive surgical repair within the CAPSNet database.

1. Materials and methods

Following CAPSNet Steering Committee and REB approval, the data set for the analysis of CDH mortality (2005–2010) was obtained. The entire cohort of CDH infants ($n=275$) was separated into the following categories: an operative group (OG; $n=240$) consisting of operative survivors (OS; $n=224$) and operative non-survivors (ONS; $n=16$), as well as those who did not receive surgery ($n=35$).

Within the group of those who did not receive surgery, infants with severe physiologic instability ($n=14$), defined as death within 48 h of birth despite maximal medical intervention, severe genetic or congenital anomalies ($n=6$), as well as those suffering from severe complications while on ECMO (i.e.) intracranial hemorrhage ($n=1$), were eliminated for the purposes of this analysis as these reasons were deemed appropriate for not offering surgical repair ($n=21$). Infants for whom care was discontinued at parental request were also excluded ($n=3$). Two of these patients could have been included in the PSC cohort while the remaining infant destabilized within 48 h of birth and fulfilled a criterion that was considered appropriate for exclusion from surgical intervention. The remaining infants ($n=11$) were designated as potential surgical candidates (PSC). Patient groupings are summarized in Fig. 1.

PSCs ($n=11$) were compared to the OG ($n=240$) and the ONS group alone ($n=16$) (Fig. 1). These comparisons included a variety of health indices, including birth weight, gestational age, umbilical cord blood pH, SNAP-II scores [4], and the presence or absence of congenital anomalies. The need for various forms of resuscitation and whether the patient underwent treatment in an ECMO center were also compared between groups. Continuous variables were analyzed using general linear models while categorical variables were analyzed using chi-square and Fisher exact tests. p Values <0.05 were considered significant.

2. Results

In the overall cohort, the median gestational age was 38 weeks with a birth weight of over 3000 g. Almost 60% of the cohort had a prenatal diagnosis. Overall survival was noted to be 82%. Forty percent of infants required either high-frequency oscillatory ventilation (HFOV) and/or inhaled nitric oxide (iNO). Two hundred forty infants underwent operative repair with the remaining 35 dying before surgery could be offered. Patch repairs were required in almost one-third of infants who had surgery and 4 patients underwent CDH repair while on ECMO. The demographic data of the entire cohort are presented in Table 1.

Data after group segregation into OG, PSC and ONS are presented in Table 2. When comparing OG and PSC groups, OG infants were less likely to have a prenatal diagnosis, HFOV, iNO or ECMO. However, PSC and ONS infants had similar rates of HFOV and iNO. Four PSC patients required ECMO compared to seven in the ONS group, a difference that was not statistically significant. Ten of 11 PSC infants were born at ECMO centers (Table 2).

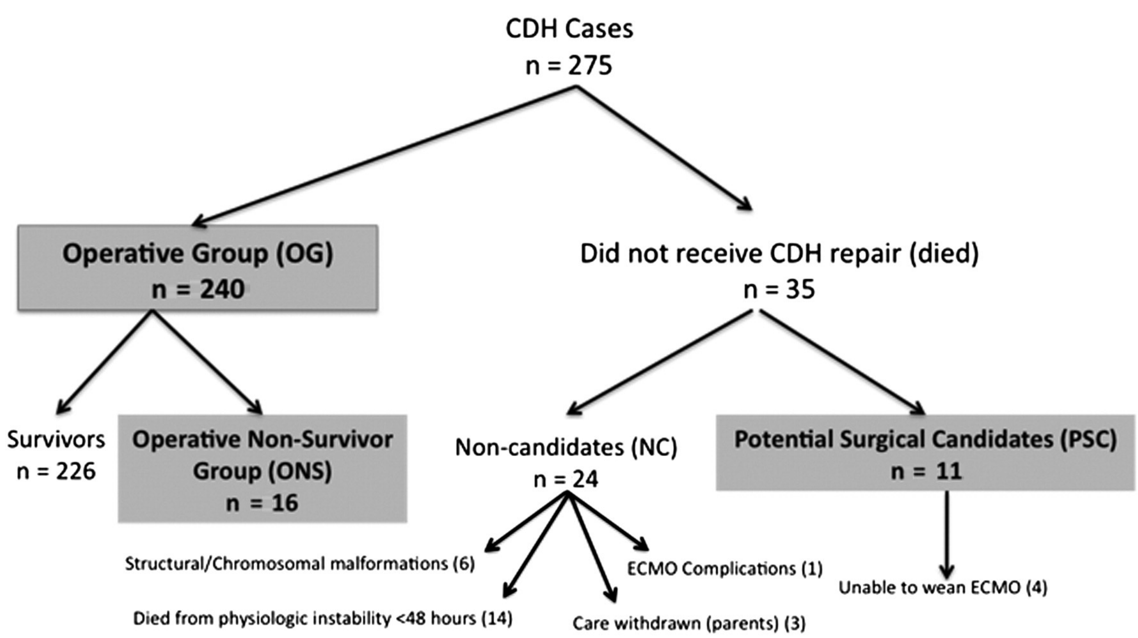

Fig. 1 Segregation of patients into an operative, potential surgical candidate and operative non-survivor groups.

Table 1 Demographic data for study population (n=275).

	n for field	Median	Range
Gestational age (weeks)	275	38	26–42
Birth weight (g)	275	3197.5	930–4930
NICU stay (days)	274	18	0–135
SNAP-II scores	209	21	5–63
	n for field	**Value (%)**	**Missing Data (%)**
Gender (male)	274	157 (57)	1 (0.4)
Liver lobe above diaphragmatic rim	198	129 (65)	77 (28)
Prenatal diagnosis	275	115 (58)	–
HFOV	275	122 (45)	–
Treatment with iNO	275	109 (40)	–
Survival	275	224 (82)	
ECMO	275	20 (7)	–
Surgical repair	275	240 (86)	–
Need for patch repair	237	80 (34)	38 (14)
Surgery on ECMO	275	4 (2)	–
Treatment in an ECMO center	275	239 (87)	–

SNAP=Score for Neonatal Acute Physiology; HFOV=high-frequency oscillatory ventilation; iNO=inhaled nitric oxide. Value indicates the number of patients with the criteria being evaluated while the number in parentheses indicates the percentage of the n for the field. Missing data indicate the number of patients for which the field was incomplete, with the percentage indicating its proportion to the total cohort of 275 infants.

Table 3 Congenital anomalies.

	n	OG (%)	PSC (%)	ONS (%)	p Value
Chromosomal abnormality	275	0 (0)	0 (0)	0 (0)	N/A
Cardiac malformations	275	19 (8) [a]	3 (27)	2 (13)	0.332
Pulmonary malformations	275	5 (2)	0 (0)	1 (6)	0.398
MSK malformations	275	3 (1) [a]	1 (9)	1 (6)	0.782
GU malformations	275	6 (3)	0 (0)	0 (0)	N/A
CNS malformations	275	3 (1)	0 (0)	0 (0)	N/A

OG=operative group; PSC=potential surgical candidate; ONS=operative non-survivor; MSK=musculoskeletal; GU=genitourinary; CNS=central nervous system.
[a] Significant difference between OG and PSC (p<0.05). p Value represents comparison between PSC and ONS.

PSCs had a median length of life of 10 days (3–18 days). With respect to congenital anomalies, there were generally no significant differences noted between the OG, PSC and ONS groups across many of the anatomical systems evaluated (Table 3). However, there were more cardiac anomalies noted in the OG group, but these likely represented minor malformations such as patent ductus arteriosus, atrial and ventricular septal defects. PSCs also did not require more resuscitative measures than those who received surgery including the need for transfusion or the requirement of inotropic support (Table 4).

The Score for Neonatal Acute Physiology II (SNAP-II) predicted mortality across the total cohort (p<0.001) (Table 2). All infants who died, including those in the ONS group, had significantly higher SNAP-II scores than those that survived (p<0.001). However, similar median SNAP-II scores were observed between PSC and ONS groups (34 vs. 30; p=0.431), indicating that illness severity, as determined by SNAP score, was inconsistently applied as a exclusion for surgery. Moreover, a threshold SNAP-II value indicating a poor surgical outcome could not be determined. In fact, more than 80% of infants with SNAP-II scores between 30 and 39 survived in the OG cohort (Fig. 2).

3. Discussion

The selection of candidates for the surgical repair of CDH has generally been predicated on the absence of associated lethal anomalies and on the physiologic stability of the infant

Table 2 Demographic and treatment data for the operative group (OG), potential surgical candidate (PSC) and operative non-survivor (ONS) group.

	n	Mean OG (SD)	Mean PSC (SD)	Mean ONS (SD)	p Value
Gestational age (weeks)	275	38 (2)	38 (2)	37 (4)	0.877
Birth weight (g)	275	3106 (641)	3071 (412)	2590 (778)	0.052
SNAP-II scores	209	19 (12) [a]	34 (12)	30 (13)	0.400
	n	**OG (%)**	**PSC (%)**	**ONS (%)**	**p Value**
Gender (male noted)	274	136 (57)	5 (45)	8 (50)	0.816
Prenatal diagnosis	275	131 (55) [b]	10 (91)	12 (75)	0.296
ECMO required	275	16 (7) [a]	4 (36)	7 (44)	0.701
HFOV	275	91 (38) [a]	10 (91)	5 (31)	0.174
Treatment with iNO	275	155 (65) [a]	10 (91)	15 (94)	0.658
Treated at ECMO center	275	206 (86)	10 (91)	12 (75)	0.895

SD=standard deviation; SNAP=Score for Neonatal Acute Physiology; HFOV=high-frequency oscillatory ventilation; iNO=inhaled nitric oxide.
[a] Significant difference between OG and PSC (p<0.001).
[b] Significant difference between OG and PSC (p<0.02). p Value represents comparison of PSC and ONS groups.

Table 4 Resuscitation data.

	n	OG (%)	PSC (%)	ONS (%)	p Value
Nasogastric tube	275	124 (52)	8 (73)	4 (25)	0.895
Fluids	275	44 (18)	4 (36)	4 (25)	0.525
Transfusion	275	1 (0.4)	0 (0)	1 (6)	0.398
CPR	275	10 (4)	0 (0)	3 (19)	0.128
Bicarbonate	275	9 (6)	1 (9)	1 (6)	0.782
Ionotropes	275	14 (6)	2 (8)	4 (25)	0.675

CPR = cardiopulmonary resuscitation. There was no difference noted between OG and PSC groups. p Values represent comparison of PSC and ONS groups.

[5,6]. While these principles make general sense, the identification of significant practice pattern variability and outcome across Canadian centers caring for these infants [3] suggests that these criteria may be inconsistently applied. Indeed, we have identified, through a CDH mortality audit of the prospective CAPSNet database involving 275 infants over a 5-year period, a subgroup of 11 infants who died without surgical repair but who, based on demographic and physiological criteria, could not be discriminated from patients who received surgery.

The overall survival rate for liveborn CDH infants has improved significantly over the last 20 years [7]. However, these reports are often based on institutional data that include only those infants who survive to operative repair, and are thus subject to case selection bias [8]. Unlike CDH patients encompassed within the concept of "hidden mortality" [9], which refers to those infants who die before transfer to a tertiary care center, there is also the clinically significant proportion of CDH infants who, after referral, die before surgical repair. These patients are often also excluded from institutional analyses but likely should not be [10]. Therefore, it has been suggested that population-based data, such as that collected by CAPSNet, provide a more accurate estimate of the true incidence and mortality of infants born with CDH since all cases in these data sets are captured at the time of prenatal diagnosis or birth. From our data, 51 (18%) of 275 infants died over the 5-year study period, of which 35 (67%) never received definitive surgical repair. This small but clinically significant proportion of patients who died before surgical repair is consistent with other reports [2,11]. While it is assumed that these infants were not provided surgical repair due to severe physiologic instability or lethal malformations, there is in fact very little known about this specific cohort. Our analysis has demonstrated that almost two-thirds of infants who died *after referral but before surgery* had significant congenital or genetic anomalies and/or died within 48 h of birth. Most importantly, however, one-third of these infants (all with isolated CDH) could not be discriminated from infants who proceeded to operative repair and thus could have been considered as potential surgical candidates (PSC). The Score for Neonatal Acute Physiology II (SNAP-II) is a validated measure that uses clinical and laboratory parameters to predict mortality in a variety of neonatal populations, including CDH [12]. Higher scores indicate more severe physiologic derangement [4]. In our study, we identified that SNAP scores for the PSC and ONS groups were similar and that the vast majority of infants in the OG who had high SNAP scores still survived after surgery. While the authors do not advocate the use of SNAP scores to determine surgical eligibility, these data underscore previous findings that there is significant practice pattern variability and that "physiological stability" is open to wide interpretation across Canada.

The concept that operative repair may not be offered to all CDH infants is controversial. While the presence of associated anomalies has been accepted as a valid reason to withhold surgery, the absolute criteria for this remain unclear. For example, Colvin et al. [10], in their Western Australian population-based CDH cohort, found that the presence of an additional single congenital malformation in liveborn infants did not affect post-natal survival. O'Mahony et al. [11] reported similar results. Within the CAPSNet database, infants with additional malformations, including congenital heart anomalies known to be associated with CDH (e.g. patent ductus arteriosus, atrial or ventricular septal defects) [13] still received surgery in the majority of cases.

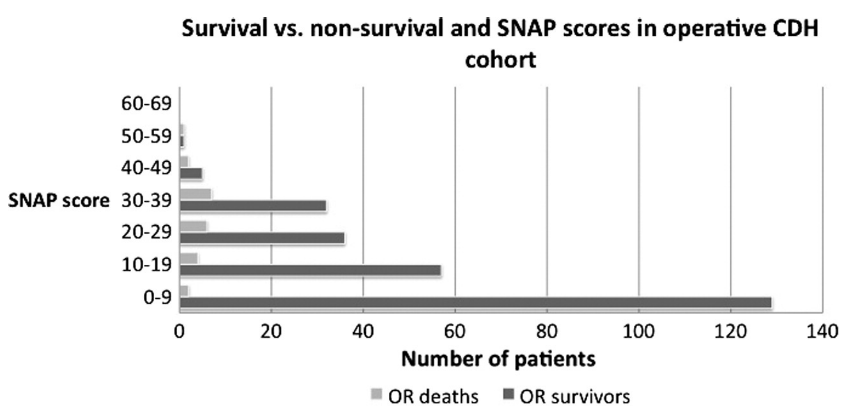

Fig. 2 SNAP-II vs. survival in the operative group cohort.

However, infants with lethal genetic syndromes identified by amniocentesis or post-natal genetic testing were precluded from surgery. Thus, it appears that minor malformations should not be considered as a valid reason to withhold surgical repair to CDH infants in Canada.

Identifying CDH patients with isolated defects for whom the offer of surgical repair is unlikely to confer survival benefit remains problematic based on our continued inability to develop standardized, evidence-based criteria in three critical areas: 1) physiologic “readiness” for repair; 2) indications for ECMO use in severe cardiorespiratory failure; and 3) timing of surgical repair once ECMO has been established. The advantage of obtaining physiologic stability prior to surgical repair has been linked to improved survival rates over the last several years [14], yet the timing of surgical repair is often debated. While general principles regarding hemodynamic parameters, ventilation pressures, oxygen needs and acid–base balance have been proposed, there are no objective criteria in the literature that clarify this issue. The timing of surgery is ultimately often left to institution or surgeon-based preferences. While attaining physiological stability makes intuitive sense, studies have failed to demonstrate a true advantage of delayed versus immediate repair [15]. Without a better definition of the criteria that indicate stability, the timing of surgery may be sub-optimal and thus lead to poorer outcomes.

The issue of physiologic stability is further clouded by the use of ECMO, which is generally reserved for those infants with severe physiologic instability and for whom the conventional therapeutic armamentarium is inadequate. While some initial studies have supported the delay of surgical repair until decannulation [16], there is emerging evidence that CDH repair while on ECMO is associated with acceptable rates of morbidity and mortality [17,18]. Proponents of this management strategy suggest that the use of new anticoagulation protocols has significantly reduced the hemorrhagic complications traditionally linked to surgery on ECMO. Furthermore, advocates suggest that ECMO may be helpful to bridge recovery after repair, especially if the repair is performed soon after cannulation and before the onset of a systemic inflammatory response that may make operative repair more difficult due to tissue edema or anasarca.

At the time of this study, ECMO for CDH was performed routinely in only four CAPSNet centers where the prevailing philosophy was to perform surgical repair after ECMO decannulation (personal communication). Interestingly, almost all PSCs were treated at ECMO centers (10/11), although ECMO utilization for PSCs was only 50% (5/10). One of these patients suffered catastrophic intra-cranial hemorrhage that precluded surgery while the other four could not be weaned, raising the question of whether acceptance of a strategy of repair on ECMO, which can be performed with acceptable morbidity [18], might have offered a survival opportunity to these infants. It is also interesting to note that five PSCs, despite being treated in an ECMO center, received neither ECMO nor surgery. Although strict criteria for ECMO initiation have not been formally established, several indices may have utility as a trigger to ECMO. These include alveolar–arterial gradients >610 mm Hg for more than 6 h, oxygenation indices greater than 40 (using pre-ductal oxygenation saturations), mean airway pressures >17 mm Hg, or persistent hypotension and/or metabolic acidosis despite maximal medical therapies that include inotropic support and nitric oxide [19–21]. While the authors are aware that the frequency of ECMO use in CDH appears to be decreasing [22,23], and that equivalent survival has been demonstrated without ECMO for infants with CDH [24,25], it is clear that the true role of ECMO in CDH must be better defined. Indeed, management decisions with the family should include an objective and open discourse regarding the risks and benefits of ECMO, including surgical repair on ECMO.

The limitations of this study are common to those associated with the analysis of any database. Specifically, the lack of autopsy data for infants prevented us from confirming the cause of death and the absence of other congenital anomalies that could have highlighted unknown differences between the PSC and ONS groups. Moreover, alternate interpretations of the data presented here are possible. Indeed, some may argue that the PSC cohort was actually “better served” than the ONS group since they avoided a painful and expensive operative procedure and complicated post-operative hospital stay. Nonetheless, unoperated patients with CDH (PSC) had a 100% mortality rate. Given the clinicians inability to discern between operated survivors (OS) and operative non-survivors (ONS) a priori, the authors feel that a reasonable default position should involve providing operative repair to all patients without identifiable exclusions.

In conclusion, our mortality audit has identified a small, but clinically significant population of patients who may have potentially benefited from surgical intervention. While it is impossible to know if surgery would have changed the outcome of these infants, it is becoming increasing clear that the criteria for *surgical ineligibility* must be more rigorously defined. Furthermore, the role and indications for ECMO must also be more clearly delineated. Consensus guidelines involving all stakeholders involved in the care of these infants are required to standardize practice patterns across Canada and to provide CDH infants with the best outcomes possible.

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